Bank Customer Analysis

Data Analytics Project — Problem Statement

1. Background

A retail bank maintains detailed records of its customers, covering demographics, occupation, income, loyalty tier, and their full financial footprint with the bank — deposits, loans, credit cards, savings, and investment-related accounts. Despite holding this rich dataset, the bank lacks a structured, data-driven view of customer value, risk, and behavior, which limits its ability to design targeted products, manage credit risk, and prioritize relationship management efforts.

2. Problem Statement

The bank needs to analyze its customer base to identify high-value customers, uncover cross-sell opportunities, flag customers at risk of loan default, and understand how value is distributed across occupations, loyalty segments, locations, and nationalities. Currently, this analysis is not performed systematically, resulting in missed opportunities for premium product targeting and delayed identification of repayment risk.

3. Objectives

- Identify top customers by bank deposits to prioritize relationship management.
- Detect high-income, low-deposit customers as targets for premium savings accounts.
- Determine which occupations contribute the most deposit value to the bank.
- Flag customers whose bank loans exceed twice their estimated income as potential repayment risks.
- Evaluate which loyalty classification segment (e.g. Jade, Gold, Silver) generates the most deposit value.
- Identify customers with high deposits but no credit card usage as cross-sell opportunities.
- Analyze deposit contribution by location to support regional strategy.
- Analyze deposit contribution by nationality to understand demographic value drivers.

4. Dataset Overview

The analysis uses a single customer-level dataset (*Banking.csv*) loaded into a MySQL database (*banking_case*) for querying. Key fields include:

- Identifiers & demographics: Client ID, Name, Age, Nationality, Occupation, Location ID, Gender, Banking Contact
- Relationship details: Joined Bank (date), Fee Structure, Loyalty Classification, Risk Weighting
- Income & wealth: Estimated Income, Superannuation Savings, Properties Owned

- Liabilities: Bank Loans, Credit Card Balance, Amount of Credit Cards
- Assets/holdings: Bank Deposits, Checking Accounts, Saving Accounts, Foreign Currency Account, Business Lending

5. Key Business Questions

- Who are our most valuable customers, and which customers should we target for premium savings accounts?
- Which occupations and loyalty segments generate the most deposit value for the bank?
- Which customers may struggle to repay their loans based on income-to-loan ratio?
- Which customers hold significant deposits but show no credit card engagement (cross-sell potential)?
- Which locations and nationalities contribute the most to total bank deposits?

6. Scope & Approach

The dataset is imported into MySQL, where SQL queries are used to segment customers, rank them by deposit value, compute risk flags (loans > 2× income), and aggregate deposits by occupation, loyalty tier, location, and nationality. A supporting Jupyter notebook is used for further exploration, visualization, and validation of these findings.

7. Expected Outcomes

- A ranked list of top-value customers for relationship management focus.
- A target list of high-income, low-deposit customers for premium savings outreach.
- A risk watch-list of customers with loan-to-income ratios above 2x.
- Deposit value breakdowns by occupation, loyalty segment, location, and nationality to guide strategy and resource allocation.